

Ali Koaik

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Education

42 Beirut - Level 6

2025 - present

42 Core Curriculum - Architecture of Digital Technologies Program

University of Science and Arts in Lebanon

2022 - present

Bachelor of Science in Computer Science

Selected Projects

Minishell / C, Unix System Calls, Process Management

September 2025

- Built a Unix-style shell from scratch using an Abstract Syntax Tree (AST) to parse and execute commands
- Created and managed processes with fork, execve, wait, and signals to support pipelines and redirections
- Implemented Bash-like features such as environment variables, quotes, expansions, built-in commands, and robust error handling

Philosophers / C, Multithreading, Synchronization

November 2025

- Built a multithreaded program in C to solve the Dining Philosophers problem
- Used one thread per philosopher and mutexes to control access to shared resources
- Prevented data races and deadlocks while ensuring correct timing and behavior

Ford-Johnson Sort Algorithm / C++, STL Containers, Algorithm Optimization

May 2025

- Implemented a merge-insertion sort algorithm minimizing comparisons through Jacobsthal-ordered insertion and bounded binary search
- Within the same module, Developed a Bitcoin exchange rate calculator using map for CSV parsing and RPN expression evaluator using stack container

Flask Web Application / Python, Flask, SQL, Bootstrap

March 2024

- Built a web application similar to Airbnb with user accounts, listings, and a search feature
- Created backend routes and templates using Flask to handle data display and user actions
- Used Bootstrap to design responsive and clean pages

Technical Skills

Languages: C, C++, Java, Python, SQL

Developer Tools: VS Code, Android Studio, Apache NetBeans

Technologies/Frameworks: Linux, GDB, Git, Bootstrap, Flask, Langchain AI Library, n8n

Project Management tools: GitHub

Relevant Coursework and Workshops

- 42 Beirut Extracurricular Cybersecurity Piscine
- 42 Beirut Extracurricular Mobile Piscine
- Participant in Lebanon's First-Ever Agentic AI Hackathon

Extracurricular Activities

Work Experience

2022 - 2024

Sales Associate & cashier | Centre Point, City Center Beirut

- Assisted customers with purchases and requests, maintaining high customer satisfaction
- Collaborated with team members to achieve daily sales targets
- Operated cash register and handled transactions accurately